

INDUSTRIAL PRODUCT DATASHEET

Wio-E5 Wireless Module (Bulk) - STM32WLE5JC, ARM Cortex-M4 and SX126x embedded, supports LoRaWAN on EU868 & US915

Ultra-compact Wio-E5 Wireless Module (Bulk) with STM32WLE5JC and SX126x, supporting LoRaWAN on EU868 & US915, built-in AT firmware, and high-performance long-range connectivity for IoT.

SKU

317990687

UPDATED

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PRICE

\$6.49(Excl. VAT)



Overview



Bulk
Package (1pcs)

The Wio-E5 Wireless Module (Bulk) is an ultra-compact, low-power LoRaWAN module based on STM32WLE5JC and embedded SX126x, supporting LoRaWAN on EU868 and US915 with (G)FSK and Long Range. It offers built-in AT command firmware, high-performance RF (up to 22dBm @868/915MHz; -136.5dBm sensitivity for SF12 with 125KHz BW), a 158dB link budget, and FCC/CE/IC/Telec/WPC/RCM certifications for low-power wide-area IoT applications.

Key Features

Note

Our Wio-E5 LoRa module is now IC certified for Canada! Currently shipped Wio-E5 modules may come with either new or old labels, but product functionality remains consistent, and certification is u

There are 100 pieces of this product available in the DE warehouse. Please note that purchases must be made as a whole unit (100 pieces) and cannot be split. To place an order, contact order@seeed.cc.

Note

In the antenna selection for Wio-E5, we provide the following two references to ensure improved RF performance and compliance with CE/FCC regulations:

1. Dipole Antenna (up to 2.0 dBi): This antenna is utilized for Wio-E5 certification. For more information, please refer to the certification report or [SKU 113070002](#).
2. Spring-loaded Antenna (up to 1.1dBi): This is the antenna employed by the [SenseCAP S210X LoRaWAN sensor](#).

This ultra-compact module has extremely low power consumption and is capable of supporting both (G)FSK and Long Range. Its built-in AT command firmware, makes it easy for developers to build their prototypes or applications. It can easily support LoRaWAN® protocols and a variety of frequency plans, making it an ideal solution for Long Range sensor nodes and a range of wireless communication applications. Whether you need to set up Wireless Meter Reading, IoT Nodes, or a Sensor Network, the Wio-E5 is perfectly suited for low power wide area IoT scenarios.

With this module, you can connect to different LoRaWAN server platforms like TheThingsStack, Chirpstack, and others.

Note

If you have a large quantity request or need any further information on Seeed IIoT products and services, please don't hesitate to contact us directly at order@seeed.cc or click [here](#). We'll be happy to offer you our best discount price. We are excited to see you can scale up and expand your business with our products and services.

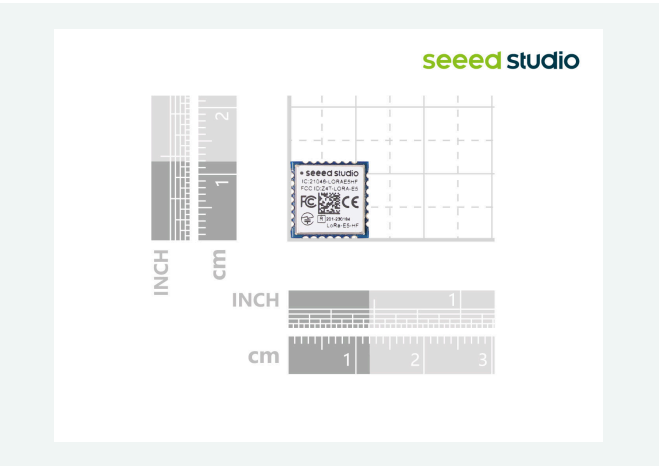
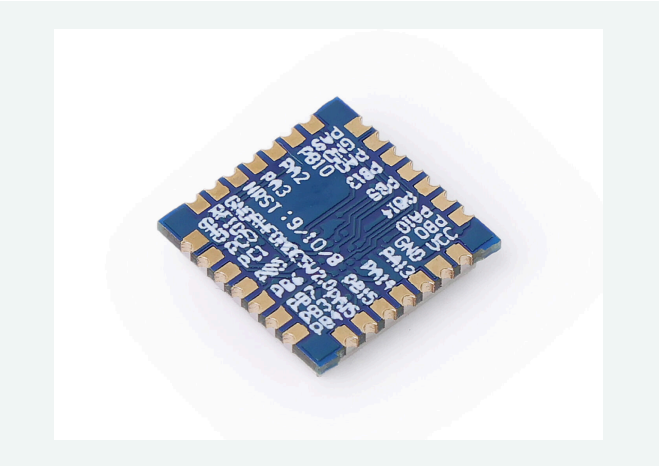
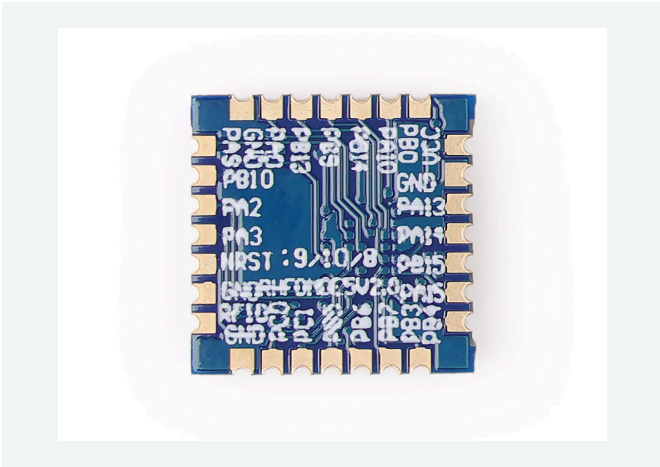
- **High Performance:** The transmit power up to 22dBm @868/915MHz; -136.5dBm sensitivity for SF12 with 125KHz BW.
- **Long Distance Use:** 158dB link budget.
- **Wireless Connectivity:** Embedded Long Range protocol, supports global Long Range frequency plan.
- **Abundant peripherals:** Equipped with a variety of GPIO interfaces like UART, I2C, and ADC, this module offers great flexibility for developers looking to expand peripherals.
- **Easy to Use:** Has built-in AT command firmware and supports custom product development using an SDK.
- **FCC, CE, IC, Telec, WPC and RCM Certified**

Note

- If your primary goal is to minimize power consumption, and your operating range is relatively short, then [Wio-E5-LE](#) is the right choice.
- Should you prioritize transmission distance and be less concerned about power consumption, the Wio-E5 would be an ideal choice. Conversely, if minimizing power consumption is your primary objective and your operational range is relatively short, the [Wio-E5-LE](#) would be a more fitting selection.

We've enhanced our development platform by updating the integrated [Wio-WM1110 Module](#), built on LoRaWAN® communication. This module offers not just the LoRa communication technology found in the Wio-E5 but also includes a radio front-end with multiple positioning features, such as GNSS (GPS/ BeiDou), Wi-Fi, and Bluetooth. It's ideal for both indoor and outdoor positioning needs of developers.

Product Gallery



Specification

Items	Parameter	Specifications	Unit
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Core	MCU	ARM Cortex M4				KB
	Storage	256KB Flash, 64KB SRAM				
Structure	Size	12(W)*12(L)*2.5(H)				mm
	Package	28 pins, SMT				
Electrical Characteristics	power supply	3.3V type				V
	Minimum sleep current	2.1uA(SOC)				uA
	Operation current (Transmitter+MCU)	50mA @10dBm in 434MHz type				mA
		111mA @22dBm in 470MHz type				
		111mA @22dBm in 868MHz type				
	Operation current (Receiver+MCU)	6.7mA @BW125kHz, 868MHz type				mA
		6.7mA @BW125kHz, 434MHz type				
		6.7mA @BW125kHz, 470MHz type				
	Output power	10dBm max @434MHz				dBm
		22dBm max @470MHz				
		22dBm max @868MHz				
	Sensitivity	@SF12, BW125kHz				dBm
		Fr(MHz)	min	typ	max	
		434	-	-13.4.5	-13.6	
		470	-	-13.6.5	-13.7.5	
868		-	-13.5	-13.7		
Harmonics	< -36dBm below 1GHz				dBm	
	< -40dBm above 1GHz				dBm	
Interface	RFIO	RF port				
	UART	3 group of UART, include 2 pins				
	I2C	1 group of I2C, include 2 pins				
	ADC	1 ADC Input, include 1 pins, 12-bit 1Msps				
	NRST	Manual reset pin input				
	SPI	1 group of SPI, include 4 pins				
Working Temperature	-40℃ ~ 85℃					

Hardware Overview



Wio-E5 (STM32WLE5JC)



RFM95 and RFM95W

Core	32-bit Arm Cortex-M4 CPU, up to 48MHz	NONE
LoRaWAN stack	Built-in with AT Command Firmware; Program with STM32Cube MCU Package	NONE
Package	12*12mm, 28 pins SMD	16*16mm, 16 pins SMD
Interfaces	UART*3, I2C*1, ADC(12-bit)*1, SPI*1, GPIO*6	SPI*1, DIO*6
Sensitivity	-116.5dBm(SF5), -121.5dBm(SF7), -136dBm(SF12)	-111dBm ~ -148dBm
Modulation	LoRa, (G)FSK, (G)MSK and BPSK	LoRa, (G)FSK, (G)MSK and OOK
Certificate	FCC, CE, IC and TELEC	NONE
Power Supply	1.8V - 3.6V	1.8V - 3.7V
RF Output Power	up to +20.8 dBm at 3.3V	up to +20 dBm

	Wio-E5	Wio-E5-LE
		
High-Performance	The transmit power up to 22dBm , ideal for long-range transmission. @868/915MHz	The transmit power is 14dBm , for slightly shorter transmission distances than Wio-E5. @868/915MHz
Low Power Consumption	111mA in transmit mode @868MHz	26mA in transmit mode @868MHz, a reduction of 76% .
Certification	FCC、CE、IC ID、TELEC、WPC、RCM	FCC、CE

Applications

- Solar-Powered Outdoor LoRaWAN Air Quality Monitor
- Low-Power Snow Depth Sensor Monitoring for Smart City Based on Wio-E5 in Norway
- Stream Runoff Monitoring Sensors with Wio-E5 for Temperature, Quality and Flow
- Smart City: Wio-E5-based Indoor Air Pollutant Detection Project
- Wio-E5 & ESP32 Powered WiLoBlino
- Loko: Tiniest GPS Tracker with Built-In Wio-E5 Module for Wild Animals, Drones

Compliance & Logistics



HSCODE	8517709000
USHSCODE	8517790000
UPC	
EUHSCODE	8517180000
COO	CHINA

FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Resource Links

- <https://www.seeedstudio.com/LoRa-E5-Wireless-Module-p-4745.html>
- [Wio-E5 datasheet and specifications](#)
- [Wio-E5 AT Command Specification](#)
- [STM32WLE5JC Datasheet](#)
- [Wio-E5-HF Certification CE-VOC-RED](#)
- [Wio-E5-HF FCC Certification -DSS](#)
- [Wio-E5-HF FCC Certification -DTS](#)
- [Wio-E5-HF TELEC Certification](#)
- [Wio-E5-HF IC Certification](#)
- [Wio-E5 kicad library](#)
- [Wio-E5 HF 3D File](#)
- [Wio-E5 Eagle Library](#)
- [STM32Cube MCU Package for STM32WL series](#)

Seeed Technology Co.,Ltd.

9F, Building G3, International E City,
Zhongshanyuan Road, Nanshan, Shenzhen, China

Tel: +86 0755-80695676

Web: www.seeed.cc

Shop: www.seeedstudio.com